

```
ubuntu@ubuntu:~$ ls -l
total 44
-rw-r--r-- 1 ubuntu ubuntu 0 Apr 9 09:36 123
-rw-r--r-- 1 ubuntu ubuntu 0 Apr 9 09:41 321
```

- d) **rm:** This command is used to remove files and directories. To remove directories one can use the `-r` option and `-I` will help with being interactive.
- e) **mv:** This command is used to move files and directories across and within the file system.

Listing files and permissions, and masking file permissions

To make things easy, I have created the following tables.

- a) *A list of the types of files.* In any of the directories (`/root` is the best example), if you execute the `'ls -l'` command, it will display the output as shown below. The first permission character depicts the type of file.

Character displayed	Type of file
-	Regular file
d	Directory file
c	Character file
b	Block file
l	Symbolic link file (explained later)
S or s	Socket file
m	Shared memory (only in certain flavours of UNIX/Linux) file
P or F	Named pipe or FIFO

```
root@ubuntu:/# cd dev
root@ubuntu:/dev# ls -l |more
total 0
crw----- 1 root root 10, 175 Apr 9 09:07 agpgart
crw----- 1 root root 10, 235 Apr 9 09:07 autofs
drwxr-xr-x 2 root root 260 Apr 9 09:07 block
drwxr-xr-x 2 root root 80 Apr 9 09:07 bsg
crw----- 1 root root 10, 234 Apr 9 09:07 btrfs-

control
drwxr-xr-x 3 root root 60 Apr 9 09:07 bus
lrwxrwxrwx 1 root root 3 Apr 9 09:07 cdrom -> sr0
lrwxrwxrwx 1 root root 3 Apr 9 09:07 cdrw -> sr0
drwxr-xr-x 2 root root 3620 Apr 9 09:08 char
crw----- 1 root root 5, 1 Apr 9 09:07 console
lrwxrwxrwx 1 root root 11 Apr 9 09:07 core -> /
proc/kcore
crw----- 1 root root 10, 59 Apr 9 09:07 cpu_dma_
latency
crw----- 1 root root 10, 203 Apr 9 09:07 cuse
```

File permissions: - {(Octal number) Read π 4 :: Write π 2 :: Execute π 1 }

```
ubuntu@ubuntu:~$ ls -la
total 108
drwxr-xr-x 1 ubuntu ubuntu 4096 Apr 11 08:47 .
drwxr-xr-x 3 root root 4096 Mar 24 09:42 ..
-rw-r--r-- 1 ubuntu ubuntu 0 Apr 9 09:36 123
-rw-r--r-- 1 ubuntu ubuntu 0 Apr 9 09:41 321
```

```
ubuntu@ubuntu:~$ chmod 0700 123
ubuntu@ubuntu:~$ ls -la
total 108
drwxr-xr-x 14 ubuntu ubuntu 4096 Apr 11 08:47 .
drwxr-xr-x 3 root root 4096 Mar 24 09:42 ..
-rwx----- 1 ubuntu ubuntu 0 Apr 9 09:36 123
```

```
ubuntu@ubuntu:~$ chmod 0777 123
ubuntu@ubuntu:~$ ls -la
total 108
drwxr-xr-x 14 ubuntu ubuntu 4096 Apr 11 08:47 .
drwxr-xr-x 3 root root 4096 Mar 24 09:42 ..
-rwxrwxrwx 1 ubuntu ubuntu 0 Apr 9 09:36 123
```

 **Note: 1)** Other possible 'chmod' options are given below:

```
chmod[who] [+/-/I][permissions] <file name>
chmod go+r go-w <file name>
```

- 2) For masking the file permissions, *umask* is used (the default value of *umask* is '0022').

Removing files forcibly

The command `'rm -f'` is used to forcibly remove files.

The *pwd*, *rmdir* and *mkdir* commands

The command `'pwd'` is used to know more about the present working directory. The commands `'mkdir'` and `'rmdir'` are used for creating and removing directories.

 **Note:** The command `'rmdir -p <Dir-Name>'` is used for removing the leaf directory. We have to include the complete path.

Disk space

The commands `'df'` (disk space) and `'du'` (disk utilisation) are mainly used to find out the disk usage.

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